vii. Origin of the Earth

*Earth is one of the planets in the solar system. Its origin is not different from that of the other planets. Different scholars have propounded different theories regarding the subject. *One of the earliest works on the subject was presented by **Immanuel Kant**. He propounded the theory of nebulous material. * In 1755 A.D. Immanuel Kant presented the Gaseous hypothesis. Later in 1796 A.D., **Laplace** presented his research on the topic which came to be known as **Nebular Hypothesis**. In 1905 A.D. **Chamberlin** and **Moulton** proposed the **Chamberlin-Moulton planetesimal hypothesis** to describe the formation of the solar system. The theory was based on the idea that a star passed close enough to the sun early in its life to cause tidal bulges to form on its surface, causing the material to be ejected repeatedly from the sun. Due to the gravitational effects of the passing star, materials would cool and condense into numerous small bodies, which they called planets. *In 1919 the famous British scientist **James Jeans** propounded the **Tidal Hypothesis** and another scientist **Sir Harold Jeffery** made some corrections to it. After some years, the theory of a wandering star around the sun started getting acceptance. The theory was called **Binary Theories**. *In 1943, Otto Schmidt propounded the interstellar dust hypothesis. He proposed that the sun, in its present form, passed through a dense interstellar cloud and emerged enveloped in a cloud of dust and gas. **Otto Schmidt** of Russia and **Carl Weizsacker** of Germany revised the Nebular Hypothesis.

They believed dust like 'Solar Nebula' with heavy helium and hydrogen surrounds the sun. A disk-shaped cloud was created by the friction of particles in the solar Nebula. The planet was formed due to the accretion process.

*In modern times the most accepted theory regarding the origin of Universe is the **Big Bang Theory** also known as **expanding Universe Theory**. Universal expansion was proved by **Sir Edwin Hubble** in 1920. According to the Big Bang Theory, a violent explosion of a tiny body occurred which resulted in huge expansion and the change of some particles into an energy form. Due to this expansion, some energy got converted into matter. This happened 13.7/13.8 billion years ago. *Another proof of this Edwin Hubble concluded that galaxies are drifting **apart**. This is important observational evidence consistent with the hypothesis of an expanding Universe. After initial expansion, the universe cooled sufficiently to allow the formation of subatomic particles and later simple atoms. Giant clouds of these elements later combined due to gravity eventually forming early stars and galaxies. The galaxies further broke down resulting in the formation of stars and stars broke down (ran out of fuel) resulting in the formation of early planets.

*In 1907 John Joly, used his knowledge of mineralogy to figure out the age of different minerals. According to him certain rock-forming minerals were often characterized by the presence of small circular dark spots or concentric rings. He called it the **Mineral Halo**. *Professor **H.H. Russell** used Uranium –Lead and Thorium-Lead to determine the age of Earth. According to it the age of our planet is 2 billion and 25 crore years and 4 billion and 60 crore year old respectively. *Radiometric dating has proven to be the most successful way of figuring out the Earth's age. As Uranium and Thorium are found in one form or the other in different quantities in every rock formation, the Uranium Dating Technique to determine the age of Earth becomes very useful.

1. Which of the following scholar has suggested that the Earth has originated from the gases and the dust particles ?

(a) James Jeans (b) H. Alfven
(c) F. Hoyle (d) O. Schmidt

I.A.S. (Pre) 1999

Ans. (d)

James Jeans and Harold Jeffreys are related to the tidal hypothesis of the origin of Earth. F. Hoyle and Littleton presented the Nova hypothesis and Otto Schmidt gave the hypothesis of gas and dust for the genesis of Earth (Interstellar dust hypothesis). According to this hypothesis, large quantities of gas and dust particles captured by the gravitational force of the sun formed the Earth.

2. Which of the following Method is used to determine the age of the Earth ?

(a) Carbon dating for the age of the fossils.
(b) Germanium dating
(c) Uranium dating
(d) All of the above

U.P.U.D.A./L.D.A. (Pre) 2006

Ans. (c)

The uranium dating method is used for determining the age of Earth.

viii. The Geological History

*Geological History of Earth follows the major events in Earth's past based on the geological time scale. The first efforts to understand the Geological History of Earth were made by a French naturalist Comte de Buffon. At present the geological time scale is used, which is a system of chronological dating that relates geological states to time. *The primary defined divisions of time are eons. Eons are divided into eras which are in turn divided into periods and epochs.

***Pre Cambrian Era** – Earth was initially molten due to extreme volcanism and frequent collisions with other bodies. Water began accumulating in the atmosphere. The only flora present at this time was seagrasses. In the warm oceans emergence of spineless creatures also took place.

A geological period is one of the several subdivisions of geological time. Every period is named after its characteristic events.

(1) **Cambrian Period** – The waters of the Cambrian period appear to have been widespread and shallow. The seas saw the emergence of creatures with spinal cord along with spinal cord during this period.

(2) **Ordovician Period** – By this time process of mountain formation had started and thus period witnessed heavy volcanic activity. Flora and fauna were only limited to the seas.

(3) **Silurian Period** – The period witnessed the emergence of terrestrial flora however leaf formation was yet not present. Diversity in the species of flora was also seen. A very prominent phenomenon of this period was large scale development and expansion of coral reefs.

(4) **Devonian Period** - The period was a time of great tectonic activity as a result of which it witnessed large scale mountain formations and volcanic activities. Various varieties of Fish emerged and evolved during this period, hence the period is also known as the 'Age of Fishes'.

(5) **Carboniferous period** – Our Earth is made up of different plates. These plates are not stationary but keep moving because of tectonic activity. Hence, these plates are responsible for shifting of the continents and oceanic floors into various parts of the planet. The Carboniferous was a time of active mountain building, as the supercontinent Pangea came together. The supercontinent was surrounded by a large water body. The sea was named Panthalassa by Wegener. The northern part of Pangea i.e. Laurasia (North America, Europe and Asia) and southern part Gondwana (South America, Africa, Madagascar, Peninsular India, Australia and Antarctica).

(6) **Permian Period** – During this period, because of geological turmoil, the birth of high mountain ranges was seen in Europe, Asia and Eastern United States of America (Appalachian). Another factor witnessed was the increase in terrestrial species.

(7) **Triassic Period** – The period witnessed the emergence of carnivorous aquatic reptiles that had a resemblance to and characteristics of fishes. The period also witnessed the first evolution of terrestrial mammals. Flies and termites also came into the picture during this period along with the origin of dinosaurs.

(8) **Jurassic Period** – This period saw the development of the size and diversity of terrestrial reptiles. Some aerial birds such as Archaeopteryx also emerged during this period. This period was dominated by dinosaurs.

Archaeological evidence of **dinosaurs** in India has been found at **Raiyoli in Mahisagar** district of Gujarat and in Bara Simla Hills near the rivers Narmada in Jabalpur district of Madhya Pradesh.

(9) **Cretaceous Period** – Geological developments and transformations were very active during this period. As a result the Rocky Mountains, Andes Mountain ranges of Europe and Isthmus of Panama were formed.

The tertiary period has been categorized into different epochs. They are as follows -

(1) **Palaeocene Epoch**

(2) **Eocene Epoch** – The epoch saw the formation of the Indian Ocean, Atlantic Ocean and the Himalayan orogeny.

(3) **Oligocene Epoch** - The epoch witnessed expansion of land mass. Geological activity continued in Europe and America and the formation of Alps started.

(4) **Miocene Epoch** – The epoch witnessed a lot of geological activity as a result of which Alps mountain range was formed, there was an uplift in the height of the lesser Himalayas and Greater Himalayas.

(5) **Pliocene Epoch** – The North Sea, Black Sea, Arabian Sea and Caspian Sea were formed the formation of the Shiwalik range also took place during this epochs.

Quaternary Period - has been divided into two Epochs- Pleistocene Epoch and Holocene Epochs.

Pleistocene Epoch - The Pleistocene Epoch is defined as the time period that began nearly 2.0 million years ago and lasted until about 10,000 years ago. The most recent ice age occurred then. Almost all the continents experienced glacier formation. The ice-sheets spread was categorized into four different times. Hence, sub-ice ages have been classified and named according to their area of occurrence. They are as follows –

Nebraskan, Kansan Illinois, Iowa and Wisconsin in North America. Similarly, Albrecht Penck and Eduard Bruckner classified the ice ages of Europe. They are as follows – Gunz, Mindel, Riss, Wurm.

*Holocene Epoch – The Holocene Epoch began 10,000 years ago.

*The melting of ice sheet led to reemergence and expansion of forests in Europe. Aquatic flora and fauna evolved into their present biological characteristics.

*The period between 1300 to 1870 A.D. is known as the Little Ice Age. During this time Europe and North America were much colder as compared to the 20th century.

1. Great Ice-Age is related to :

(a) Pleistocene (b) Oligocene

(c) Holocene (d) Eocene

M.P.P.C.S. (Pre) 2013

Ans. (a)

The Pleistocene Epoch was the time during which extensive ice sheets and other glaciers formed on the landmass. This time period has been informally referred to as the "Great Ice Age" and began approximately 2 million years ago and ended about. ten thousand years ago.

2. Which of the following periods has generally been considered to be the 'Little Ice Age'?

(a) 750 A.D. - 850 A.D.

(b) 950 A.D. - 1250 A.D.

(c) 1650 A.D. - 1870 A.D.

(d) 8000 to 10,000 years B.P. (Before Present)

U.P.P.C.S. (Pre) (Re-Exam) 2015

Ans. (c)

The Little Ice Age is a period between 1300 A.D. to 1870 A.D. during which Europe and North America were subjected to much colder winters than during the 20th century. Thus (c) is the correct option.

3. What was the period of the Dinosaurs?

(a) Five crore years ago

(b) Eighteen crores years ago

(c) Forty crores years ago

(d) Eighty crores years ago

M.P.P.C.S. (Pre) 2005

Ans. (b)

The time of existence of the Jurassic Period is estimated to be 20.8 crore years ago and ended about 14.4 crore years ago. This period is called the Age of Dinosaurs. Its remains were found at Raiyali in Mahisagar district of Gujarat and Barashimla hills of Jabalpur M.P. Thus, the correct answer is option (b).

4. Continents have drifted apart because of-

(a) Volcanic eruptions

(b) Tectonic activities

(c) Folding and faulting of rocks

(d) All of the above

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (b)

Continents rest on massive slabs of rocks called tectonic plates. These plates are always moving and interacting in a process called plate tectonic activity. Through these tectonic activities, continents shift position on Earth's lithosphere.

5. Which one of the following continents was not a part of Gondwana Land?

(a) North America (b) South America

(c) Africa (d) Australia

U.P.P.C.S. (Mains) 2016

Ans. (a)

Gondwana land included most of the landmasses of today, including Antarctica, South America, Africa, Madagascar the Australian continent as well as the Arabian Peninsula and Peninsular India. North America is not a part of Gondwana land.

6. The southern continent broken from Pangaea is called:

(a) Pacific Ocean

(b) Laurasia

(c) Gondwanaland

(d) More than one of the above

(e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (c)

See the explanation of the above question.

7. India was the part of ancient Gondwana land Supercontinent. It includes the following landmass of the present :

(a) South America (b) Africa

(c) Australia (d) All of above

M.P.P.C.S. (Pre) 2008

Ans. (d)

In the Carboniferous period, the whole landmass on the Earth was unified and was named Pangea. This Pangea supercontinent was surrounded by a superocean which was named 'Panthalassa' by Wegener. Due to the process of continental drift, Pangea split into a northernmost and southernmost supercontinent. The northernmost landmass was known as Laurasia. The southernmost supercontinent was known as Gondwanaland which split into South-America, Australia, Africa, Madagascar, Antarctica, Arabian Peninsula and Peninsular India.

8. Cocos Plate the between:

(a) Central America and Pacific Plate
(b) South America and Pacific Plate
(c) Red Sea and Persian Gulf
(d) Asiatic Plate and Pacific Plate
(e) None of the above / More than one of the above

63rd B.P.C.S. (Pre) 2017

Ans. (a)

Cocos Plate lies between Central America and Pacific Plate.

9. On the earth, originally, there was only one huge landmass which is known as –

(a) Panthalassa (b) Pangea
(c) Laurasia (d) Gondwanaland

U.P.P.C.S. (Mains) 2016

Ans. (b)

Originally, the huge landmass on Earth was unified and known as 'Pangea'. Pangaea was a supercontinent.

10. First fossil evidence for the existence of life on earth is-

(a) 0.3 million years back (b) 3 million years back
(c) 5 million years back (d) 10 million years back

U.P.P.C.S. (Mains) 2005

Ans. (*)

The first fossil evidence of the existence of life on Earth dates back to the Archean Eon approximately 3.5 billion years ago. Thus none of the options is correct.

11. Which of the following phenomena might have influenced the evolution of organisms?

1. Continental drift

2. Glacial cycles

Select the correct answer using the code given below.

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

I.A.S. (Pre) 2014

Ans. (c)

According to the book "Bhautik Bhugol Ka Swarup" written by Savindra Singh, the continental drift and glacial cycle both the phenomenon have influenced the evolution of organisms.

12. The theory that states "pieces of the Earth's crust are in constant, slow motion driven by movement in the mantle" is called:

(a) the theory of continental drift
(b) the theory of Pangaea
(c) the theory of plate tectonics
(d) the theory of plate boundaries

69th B.P.S.C. (Pre), 2023

Ans. (c)

The theory of plate tectonics states that - 'pieces of the Earth's crust are in constant, slow motion driven by movement in the mantle. According to this theory, the Earth surface is composed of several plates. Plates are the solid and rigid upper part above the asthenosphere segmented into several blocks. Those blocks are known as lithospheric plates. There are seven major plates- Eurasian, African, Indo-Australian, Pacific, North American, South American and Antarctic plates. Significantly plates constitute the entire surface of the Earth.

13. The process that continually adds new crust is:

(a) subduction
(b) earthquake
(c) seafloor spreading
(d) convection

69th B.P.S.C. (Pre), 2023

Ans. (c)

The process that continually adds new crust is seafloor spreading. Significantly, new crust is created by seafloor spreading. Old crust is destroyed by subduction. It is important to mention here that the two factors roughly balance each other. So the shape and diameter of the Earth remain unchanged. Spreading seafloors are only an aspect of plate tectonics, another is subduction.

14. Folding is the result of-

(a) Epeirogenic force (b) Coriolis force
(c) Orogenic force (d) Exogenic force

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (c)

The Earth's surface experiences different types of forces. The orogenic force takes millions of years to build a mountain from plain and sea bed. These forces due to the interaction between tectonic plates, crumple and push upward to form mountain ranges. Thus folding is the result of orogenic force.

15. Which of the following processes do not come under diastrophism?

(a) Orogenic (b) Heterogenic
(c) Earthquakes (d) Plate Tectonics

M.P.P.C.S. (Pre) 2023

Ans. (*)